

NEHITH SAI VEMULAPALLI

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Master's graduate in Robotics and AI specializing in 3D Reconstruction, SLAM, and Multi-Agent Systems. Published author with 4 peer-reviewed papers, bridging theoretical algorithm design with practical ROS2 implementation. Award-winning innovator (UCL Xplore Winner) with proven success delivering end-to-end autonomous systems—from multi-sensor fusion pipelines to deployed swarm hardware.

EDUCATION

University College London (UCL) – Marshgate, Department of Computer Science London, UK
MSc Robotics and Artificial Intelligence — Merit Sep 2024 – Dec 2025

Thesis: Hybrid 3D Reconstruction Pipeline for Cultural Heritage Preservation in Extreme Environments

Key Modules: Robot Vision & Navigation, Machine Learning for Robotics, Aerial Robotics, Soft Robotics, Estimation & Control, Computer Vision & Sensing, Motion Planning, Affective Computing

Amrita Vishwa Vidyapeetham – School of Computing Kerala, India
Bachelor of Technology in Artificial Intelligence Engineering Oct 2020 – Jun 2024
First Class with Distinction — GPA: 8.27/10

PUBLICATIONS

Peer-Reviewed Publications:

- Vemulapalli, N. S.** (2025). Strategic Network Intervention: Simulating and Blocking Contagion Spread Using Dominating Sets. *World Congress on Smart Computing*, pp. 109–122.
- Vemulapalli, N. S.** (2024). Reinforcement Learning-Based Autonomous Landing of AirSim Simulated Quadcopter in Unreal Engine. *IEEE 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*.
- Vemulapalli, N. S.** (2024). Application of Unsupervised Learning in Detecting Behavioural Patterns in E-commerce Customers. *Proceedings of the 5th International Conference on Data Science, Machine Learning and Applications (ICDSMLA)*, Volume 1, pp. 1208–1217.
- Vemulapalli, N. S.** (2023). Face Detection with Landmark using YOLOv8. *IEEE 3rd International Conference on Electronics, Future Technologies and Engineering Trends (ICEFEET)*.

Publications in Preparation:

- Vemulapalli, N. S.** Hybrid 3D Reconstruction Pipeline for Cultural Heritage Preservation in Extreme Environments. (*Target: 3DV / Heritage Science*).
- Vemulapalli, N. S.** Self-Reconfigurable Robots for Space Exploration: Morphology-Shifting Simulation and Terrain-Adaptive Locomotion. (*Target: IEEE International Conference on Robotics and Automation*).

EXPERIENCE

Graduate Research Assistant – UCL Here East London, UK | May 2025 – Dec 2025

- Conducted research on hybrid 3D reconstruction for cultural heritage preservation, combining photogrammetry, LiDAR, and neural rendering, under supervision of Dr. Vijay Pawar, Associate Professor in Digital Construction Technology at UCL Bartlett
- Collaborated with cross-functional research teams to design experiments and advance state-of-the-art in 3D reconstruction

Student Entrepreneur – AquaScout London, UK | Nov 2024 – Mar 2025

- Won UCL Xplore programme competition among 50+ teams with drone-based water quality testing proposal
- Completed 12-week Venture Builder programme, developing business model, market analysis, and go-to-market strategy
- Designed technical architecture for autonomous UAV water monitoring system with preliminary simulation validation

Student Ambassador – UCL Department of Computer Science London, UK | Dec 2024 – Sep 2025

- Organized technical talks and networking sessions with visiting faculty, researchers, and industry professionals

Research Intern – National Institute of Technology Karnataka Remote | Nov 2021 – Jan 2022

- Conducted data analysis on heart attack rates based on age and demographics across India, applying statistical methods to identify risk patterns
- Developed predictive models for cardiovascular risk assessment, gaining hands-on experience in data preprocessing, feature engineering, and model selection

PROJECTS

Hybrid 3D Reconstruction Pipeline for Cultural Heritage Preservation

- Developed hybrid pipeline combining unsynchronized photogrammetry (COLMAP) and LiDAR data for robust 3D reconstruction in extreme environments, addressing temporal and spatial misalignment challenges
- Implemented ICP algorithm for precise point cloud alignment and trained 3D Gaussian Splatting model for high-fidelity scene reconstruction with photorealistic rendering capabilities
- *Technologies:* COLMAP, Open3D, Livox Mid-360 LiDAR, 3D Gaussian Splatting, Python

Graph-Based SLAM, ORB-SLAM Evaluation & Multi-Sensor Integration

- Implemented Factor Graph SLAM with pose graph optimization, demonstrating superior performance over EKF by preventing cumulative drift in long-term navigation scenarios
- Systematically evaluated Monocular ORB-SLAM on KITTI and TUM datasets, achieving RMSE of 4.91m through feature detection optimization and loop closure refinement
- *Technologies:* ORB-SLAM, ROS2, COLMAP, EVO, Factor Graph Optimization, Python, C++

Structural Inspection Path Planning & Multi-Agent Swarm Control

- Designed autonomous inspection system using TSP optimization for efficient coverage path planning of bridge and building structures
- Implemented centralized and decentralized swarm control strategies for coordinated multi-agent navigation with collision avoidance, validated in simulation and deployed on physical Crazyflie drones
- *Technologies:* ROS2, Aerostack2, Gazebo, RViz, Crazyflie Drones, TSP Optimization, Python

Precision Manipulation & Adaptive Classification in Cluttered Environments

- Designed pick-and-place system implementing shape-specific grasp planning using geometric primitives and force closure analysis for reliable object manipulation
- Developed collision-free motion planning using RRT* algorithm with dynamic obstacle avoidance and integrated 2D/3D shape classification pipeline for adaptive grasping
- *Technologies:* Point Cloud Segmentation, Grasp Planning, RRT*, PyTorch, Open3D, Python

Self-Reconfigurable Robots for Space Exploration

- Designed morphology-shifting robotic system with adaptive locomotion modes optimized for planetary exploration across diverse terrains
- Simulated multi-modal locomotion strategies in PyBullet using URDF modeling and soft robotics principles, implementing RL controller for autonomous mode switching
- *Technologies:* PyBullet, URDF, Soft Robotics Simulation, Reinforcement Learning, Python

Visual-Inertial SLAM Rover (*In Progress*)

- Building a wheeled robot for autonomous room mapping and navigation using monocular camera and IMU, with EKF sensor fusion via `robot_localization` and ROS2 Nav2 for closed-loop path planning
- Hardware wired and calibrated: ESP32 microcontroller, MPU-6050 IMU, Logitech C270 webcam; chassis 3D printed on Bambu A1
- *Technologies:* ROS2, Nav2, ORB-SLAM3/RTAB-Map, EKF, ESP32, Python, 3D Printing

SKILLS

Programming Languages: Python (Proficient), C++ (Working Knowledge), MATLAB, LaTeX

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Aerostack2, PyBullet, Webots, Arduino, Raspberry Pi, Pinocchio

Computer Vision & SLAM: ORB-SLAM, COLMAP, OpenCV, Open3D, Object Detection, Sensor Fusion, Multi-sensor Fusion, Point Cloud Processing

3D Reconstruction & Sensing: 3D Gaussian Splatting, LiDAR Processing (Livox Mid-360), Photogrammetry, CloudCompare

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib

Development Tools: Git, Docker, Linux, Anaconda/Mamba, Fusion 360, SolidWorks, 3D Printing (Bambu A1)

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Deep Learning with PyTorch | Fundamentals of Deep Learning | Image Processing with Keras in Python | Introduction to Cloud Computing | Create Machine Learning Models

Honors: UCL Xplore Programme Winner | First Class with Distinction (B.Tech)

LANGUAGES

English (Professional Working), Telugu (Native), Hindi (Professional Working)